

12-Week Node.js Backend Development Roadmap

Week 1: Fundamentals of Backend & Node.js

- Backend concepts, Node.js architecture
- Installing Node.js and npm
- Core modules (fs, path, os)
- Blocking vs non-blocking I/O
- Practice: CLI tool and EventEmitter

Week 2: Asynchronous JS in Node

- Callbacks, Promises, async/await
- Error handling with try/catch
- Practice: Promise to async/await conversion

Week 3: HTTP Module & Creating Servers

- Native http module
- Manual routing
- Serving HTML and JSON
- Practice: Simple web server

Week 4: Express.js Framework

- Routing and middleware
- Static files
- MVC pattern
- Practice: Blog API

Week 5: RESTful API & Postman

- REST architecture
- Status codes
- Practice: To-do API, Postman testing

Week 6: MongoDB & Mongoose

- CRUD with Mongoose
- Schema design
- Relationships (populate)

- Practice: User-product model

Week 7: Authentication

- bcrypt, JWT
- Middleware auth
- Practice: Signup/login API with protected routes

Week 8: File Uploads & Emails

- Multer, Nodemailer
- Practice: Profile pic upload, email notification

Week 9: Testing

- Jest, Supertest
- Mocks and coverage
- Practice: API tests

Week 10: Security & Validation

- Helmet.js, rate limiting
- Joi/Zod validation
- Practice: Secure and validate API

Week 11: Real-Time & Background Jobs

- Socket.io, Bull & Redis
- Cron jobs
- Practice: Chat or notification system

Week 12: Deployment & Final Project

- Docker, environment setup
- CI/CD with GitHub Actions
- Practice: Full-stack project and deploy